

Unit 111 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 55.7 ac, BAF 20 variable-radius cruise on a 300 ft grid, 27 plots, 300 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (SE 3.5 %)	Type	Sierran mixed conifer
Trees per acre	110	CWHR size / density	4 / D (cover 86 %)
QMD	19.2 in	SDI	296 (51 % of max 586)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	216 sq ft/ac 20.9 in	BA QMD	7 sq ft/ac 7.9 in
TPA	91	TPA	20
Volume	43.3 MBF/ac (7882 cu ft)	Volume	2.9 green tons/ac (96 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~154 sq ft/ac BA	Sampling error	7.1 % at 95 % (t = 2.06, n = 27)
Species (BA)	WF 71, PP 63, DF 36, IC 35, SP 18	Stratum standard	40 % - MEETS; 27 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	12.6	3.7	52	0.3
8-11.9	13	17.1	9.6	170	0.9
12-15.9	36	25.6	26.7	676	3.7
16-19.9	46	19.8	34.1	1051	5.8
20-23.9	49	14.1	36.3	1244	6.8
24-27.9	43	8.7	31.9	1188	6.5
28-31.9	38	5.9	28.1	1154	6.3
32-35.9	25	3.0	18.5	783	4.3
36-39.9	16	1.5	11.9	595	3.3
40-43.9	11	0.9	8.1	389	2.1
44-47.9	13	0.8	9.6	477	2.6
48-51.9	2	0.1	1.5	72	0.4
52-55.9	2	0.1	1.5	79	0.4
60-63.9	1	0.0	0.7	47	0.3
All	300	110.2	222.2	7977	43.9

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	96	34.7	71.1	19.4	2572	14.1
PP	85	32.9	63.0	18.7	2225	12.2
DF	48	16.5	35.6	19.9	1235	6.8
IC	47	15.8	34.8	20.1	1342	7.4
SP	24	10.4	17.8	17.7	604	3.3
All	300	110.2	222.2	19.2	7977	43.9

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 111: plots on the 300 ft grid (orange = QA flag)

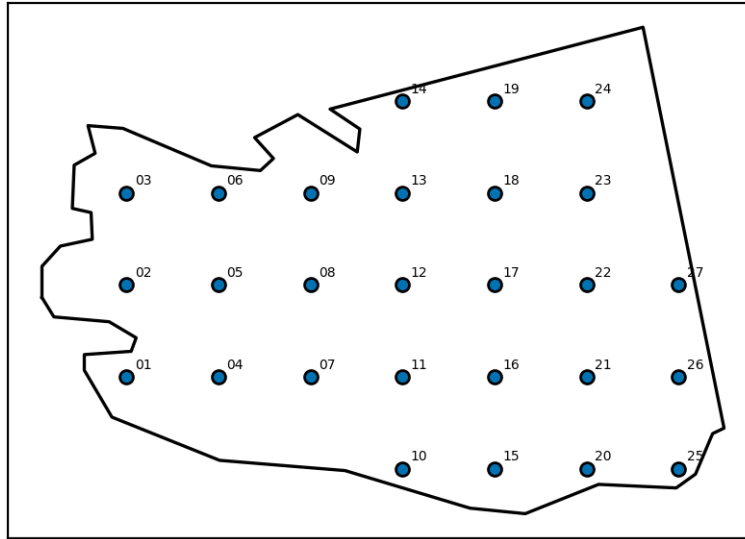


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
111-01	13	260	94	ok
111-02	11	220	49	ok
111-03	11	220	64	ok
111-04	9	180	60	ok
111-05	14	280	22	ok
111-06	10	200	25	ok
111-07	13	260	22	ok
111-08	9	180	30	ok
111-09	11	220	106	ok
111-10	8	160	99	ok
111-11	13	260	56	ok
111-12	14	280	65	ok
111-13	12	240	124	ok
111-14	10	200	122	ok
111-15	9	180	16	ok
111-16	11	220	86	ok
111-17	13	260	12	ok
111-18	13	260	32	ok
111-19	10	200	112	ok
111-20	9	180	127	ok
111-21	10	200	18	ok
111-22	12	240	62	ok
111-23	12	240	46	ok
111-24	12	240	62	ok
111-25	6	120	10	ok
111-26	11	220	75	ok
111-27	14	280	103	ok